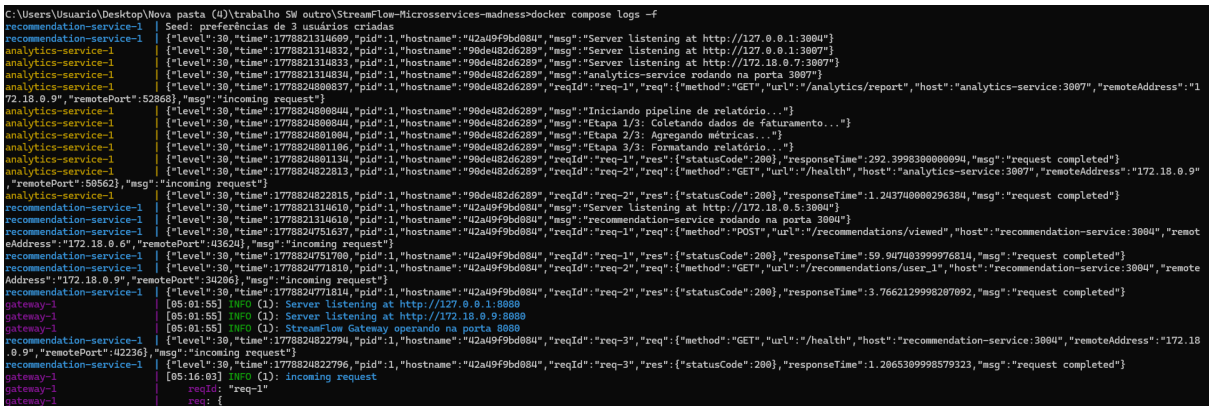
# StreamFlow — Guia de Debugging

Este guia ajuda a diagnosticar problemas e entender o comportamento do sistema.

## Logs

### Ver logs de todos os serviços

### docker compose logs -f



```
C:\Users\Usuario\Desktop\Nova pasta (4)\trabalho SW outro\StreamFlow-Microsservices-madness>docker compose logs -f
recommendation-service-1  | Seed: preferências de 3 usuários criadas
recommendation-service-1  | {"level":30,"time":1778821314609,"pid":1,"hostname":"42a49f9bd084","msg":"Server listening at http://127.0.0.1:3004"}
analytics-service-1       | {"level":30,"time":1778821314832,"pid":1,"hostname":"90de482d6289","msg":"Server listening at http://127.0.0.1:3007"}
analytics-service-1       | {"level":30,"time":1778821314833,"pid":1,"hostname":"90de482d6289","msg":"Server listening at http://172.18.0.7:3007"}
analytics-service-1       | {"level":30,"time":1778821314834,"pid":1,"hostname":"90de482d6289","msg":"analytics-service rodando na porta 3007"}
analytics-service-1       | {"level":30,"time":1778824800837,"pid":1,"hostname":"90de482d6289","reqId":"req-1","req":{"method":"GET","url":"/analytics/report","host":"analytics-service:3007","remoteAddress":"1
72.18.0.9","remotePort":52868},"msg":"incoming request"}
analytics-service-1       | {"level":30,"time":1778824800844,"pid":1,"hostname":"90de482d6289","msg":"Iniciando pipeline de relatório..."}
analytics-service-1       | {"level":30,"time":1778824800844,"pid":1,"hostname":"90de482d6289","msg":"Etapa 1/3: Coletando dados de faturamento..."}
analytics-service-1       | {"level":30,"time":1778824801004,"pid":1,"hostname":"90de482d6289","msg":"Etapa 2/3: Agregando métricas..."}
analytics-service-1       | {"level":30,"time":1778824801106,"pid":1,"hostname":"90de482d6289","msg":"Etapa 3/3: Formatando relatório..."}
analytics-service-1       | {"level":30,"time":1778824801134,"pid":1,"hostname":"90de482d6289","reqId":"req-1","res":{"statusCode":200},"responseTime":292.399830000094,"msg":"request completed"}
analytics-service-1       | {"level":30,"time":1778824822813,"pid":1,"hostname":"90de482d6289","reqId":"req-2","req":{"method":"GET","url":"/health","host":"analytics-service:3007","remoteAddress":"172.18.0.9"
,"remotePort":50562},"msg":"incoming request"}
analytics-service-1       | {"level":30,"time":1778824822815,"pid":1,"hostname":"90de482d6289","reqId":"req-2","res":{"statusCode":200},"responseTime":1.243740000296384,"msg":"request completed"}
recommendation-service-1  | {"level":30,"time":1778821314610,"pid":1,"hostname":"42a49f9bd084","msg":"Server listening at http://172.18.0.5:3004"}
recommendation-service-1  | {"level":30,"time":1778821314610,"pid":1,"hostname":"42a49f9bd084","msg":"recommendation-service rodando na porta 3004"}
recommendation-service-1  | {"level":30,"time":1778824751637,"pid":1,"hostname":"42a49f9bd084","reqId":"req-1","req":{"method":"POST","url":"/recommendations/viewed","host":"recommendation-service:3004","remot
eAddress":"172.18.0.6","remotePort":43624},"msg":"incoming request"}
recommendation-service-1  | {"level":30,"time":1778824751700,"pid":1,"hostname":"42a49f9bd084","reqId":"req-1","res":{"statusCode":200},"responseTime":59.947403999976814,"msg":"request completed"}
recommendation-service-1  | {"level":30,"time":1778824771810,"pid":1,"hostname":"42a49f9bd084","reqId":"req-2","req":{"method":"GET","url":"/recommendations/user_1","host":"recommendation-service:3004","remote
Address":"172.18.0.9","remotePort":34206},"msg":"incoming request"}
recommendation-service-1  | {"level":30,"time":1778824771814,"pid":1,"hostname":"42a49f9bd084","reqId":"req-2","res":{"statusCode":200},"responseTime":3.7662129998207092,"msg":"request completed"}
gateway-1                 | [05:01:55] INFO (1): Server listening at http://127.0.0.1:8080
gateway-1                 | [05:01:55] INFO (1): Server listening at http://172.18.0.9:8080
gateway-1                 | [05:01:55] INFO (1): StreamFlow Gateway operando na porta 8080
recommendation-service-1  | {"level":30,"time":1778824822794,"pid":1,"hostname":"42a49f9bd084","reqId":"req-3","req":{"method":"GET","url":"/health","host":"recommendation-service:3004","remoteAddress":"172.18
.0.9","remotePort":42236},"msg":"incoming request"}
recommendation-service-1  | {"level":30,"time":1778824822796,"pid":1,"hostname":"42a49f9bd084","reqId":"req-3","res":{"statusCode":200},"responseTime":1.206530999857932 3,"msg":"request completed"}
gateway-1                 | [05:16:03] INFO (1): incoming request
gateway-1                 |     reqId: "req-1"
gateway-1                 |     req: {
```

C:\Users\Usuario\Desktop\Nova pasta (4)\trabalho SW outro\StreamFlow-Microsservices-madness>docker compose logs -f
recommendation-service-1  | Seed: preferências de 3 usuários criadas
recommendation-service-1  | {"level":30,"time":1778821314609,"pid":1,"hostname":"42a49f9bd084","msg":"Server listening at http://127.0.0.1:3004"}
analytics-service-1  | {"level":30,"time":1778821314832,"pid":1,"hostname":"90de482d6289","msg":"Server listening at http://127.0.0.1:3007"}
analytics-service-1  | {"level":30,"time":1778821314833,"pid":1,"hostname":"90de482d6289","msg":"Server listening at http://172.18.0.7:3007"}
analytics-service-1  | {"level":30,"time":1778821314834,"pid":1,"hostname":"90de482d6289","msg":"analytics-service rodando na porta 3007"}
analytics-service-1  | {"level":30,"time":1778824800837,"pid":1,"hostname":"90de482d6289","reqId":"req-1","req":{"method":"GET","url":"/analytics/report","host":"analytics-service:3007","remoteAddress":"172.18.0.9","remotePort":52868},"msg":"incoming request"}

analytics-service-1 |
{"level":30,"time":1778824800844,"pid":1,"hostname":"90de482d6289","msg":"Iniciando pipeline de relatório..."}
analytics-service-1 |
{"level":30,"time":1778824800844,"pid":1,"hostname":"90de482d6289","msg":"Etapa 1/3: Coletando dados de faturamento..."}
analytics-service-1 |
{"level":30,"time":1778824801004,"pid":1,"hostname":"90de482d6289","msg":"Etapa 2/3: Agregando métricas..."}
analytics-service-1 |
{"level":30,"time":1778824801106,"pid":1,"hostname":"90de482d6289","msg":"Etapa 3/3: Formatando relatório..."}
analytics-service-1 |
{"level":30,"time":1778824801134,"pid":1,"hostname":"90de482d6289","reqId":"req-1","res":{"statusCode":200},"responseTime":292.3998300000094,"msg":"request completed"}
analytics-service-1 |
{"level":30,"time":1778824822813,"pid":1,"hostname":"90de482d6289","reqId":"req-2","req":{"method":"GET","url":"/health","host":"analytics-service:3007","remoteAddress":"172.18.0.9","remotePort":50562},"msg":"incoming request"}
analytics-service-1 |
{"level":30,"time":1778824822815,"pid":1,"hostname":"90de482d6289","reqId":"req-2","res":{"statusCode":200},"responseTime":1.243740000296384,"msg":"request completed"}
recommendation-service-1 |
{"level":30,"time":1778821314610,"pid":1,"hostname":"42a49f9bd084","msg":"Server listening at http://172.18.0.5:3004"}
recommendation-service-1 |
{"level":30,"time":1778821314610,"pid":1,"hostname":"42a49f9bd084","msg":"recommendation-service rodando na porta 3004"}
recommendation-service-1 |
{"level":30,"time":1778824751637,"pid":1,"hostname":"42a49f9bd084","reqId":"req-1","req":{"method":"POST","url":"/recommendations/viewed","host":"recommendation-service:3004","remoteAddress":"172.18.0.6","remotePort":43624},"msg":"incoming request"}
recommendation-service-1 |
{"level":30,"time":1778824751700,"pid":1,"hostname":"42a49f9bd084","reqId":"req-1","res":{"statusCode":200},"responseTime":59.947403999976814,"msg":"request completed"}
recommendation-service-1 |
{"level":30,"time":1778824771810,"pid":1,"hostname":"42a49f9bd084","reqId":"req-2","req":{"method":"GET","url":"/recommendations/user_1","host":"recommendation-service:3004","remoteAddress":"172.18.0.9","remotePort":34206},"msg":"incoming request"}
recommendation-service-1 |
{"level":30,"time":1778824771814,"pid":1,"hostname":"42a49f9bd084","reqId":"req-2","res":{"statusCode":200},"responseTime":3.7662129998207092,"msg":"request completed"}
gateway-1 | [05:01:55] INFO (1): Server listening at http://127.0.0.1:8080
gateway-1 | [05:01:55] INFO (1): Server listening at http://172.18.0.9:8080
gateway-1 | [05:01:55] INFO (1): StreamFlow Gateway operando na porta 8080
recommendation-service-1 |
{"level":30,"time":1778824822794,"pid":1,"hostname":"42a49f9bd084","reqId":"req-3","req":{"

method":"GET","url":"/health","host":"recommendation-service:3004","remoteAddress":"172.18.0.9","remotePort":42236},"msg":"incoming request"}
recommendation-service-1 |
{"level":30,"time":1778824822796,"pid":1,"hostname":"42a49f9bd084","reqId":"req-3","res":{"statusCode":200},"responseTime":1.2065309998579323,"msg":"request completed"}
gateway-1 | [05:16:03] INFO (1): incoming request
gateway-1 | reqId: "req-1"
gateway-1 | req: {
gateway-1 | "method": "GET",
gateway-1 | "url": "/",
gateway-1 | "host": "localhost:8080",
gateway-1 | "remoteAddress": "172.18.0.1",
gateway-1 | "remotePort": 59132
gateway-1 | }
gateway-1 | [05:16:03] INFO (1): Route GET:/ not found
gateway-1 | reqId: "req-1"
gateway-1 | [05:16:03] INFO (1): request completed
gateway-1 | reqId: "req-1"
gateway-1 | res: {
gateway-1 | "statusCode": 404
gateway-1 | }
gateway-1 | responseTime: 25.292331999866292
gateway-1 | [05:16:03] INFO (1): incoming request
gateway-1 | reqId: "req-2"
gateway-1 | req: {
gateway-1 | "method": "GET",
gateway-1 | "url": "/favicon.ico",
gateway-1 | "host": "localhost:8080",
gateway-1 | "remoteAddress": "172.18.0.1",
billing-service-1 | Seed: assinaturas e faturas criadas
billing-service-1 |
{"level":30,"time":1778821314716,"pid":1,"hostname":"c98f86128e30","msg":"Server listening at http://127.0.0.1:3006"}
billing-service-1 |
{"level":30,"time":1778821314717,"pid":1,"hostname":"c98f86128e30","msg":"Server listening at http://172.18.0.4:3006"}
billing-service-1 |
{"level":30,"time":1778821314717,"pid":1,"hostname":"c98f86128e30","msg":"billing-service rodando na porta 3006"}
billing-service-1 |
{"level":30,"time":1778824787072,"pid":1,"hostname":"c98f86128e30","reqId":"req-1","req":{"method":"GET","url":"/billing/user_1","host":"billing-service:3006","remoteAddress":"172.18.0.9","remotePort":46140},"msg":"incoming request"}
billing-service-1 |
{"level":30,"time":1778824787096,"pid":1,"hostname":"c98f86128e30","reqId":"req-1","res":{"statusCode":200},"responseTime":20.090739000122994,"msg":"request completed"}
billing-service-1 |
{"level":30,"time":1778824822804,"pid":1,"hostname":"c98f86128e30","reqId":"req-2","req":{"

method":"GET","url":"/health","host":"billing-service:3006","remoteAddress":"172.18.0.9","remotePort":42272},"msg":"incoming request"}
billing-service-1 |
{"level":30,"time":1778824822805,"pid":1,"hostname":"c98f86128e30","reqId":"req-2","res":{"statusCode":200},"responseTime":0.65776800038293,"msg":"request completed"}
catalog-service-1 | Seed: 10 filmes/séries inseridos
catalog-service-1 |
{"level":30,"time":1778821314898,"pid":1,"hostname":"6fe30a59ec47","msg":"Server listening at http://127.0.0.1:3002"}
catalog-service-1 |
{"level":30,"time":1778821314899,"pid":1,"hostname":"6fe30a59ec47","msg":"Server listening at http://172.18.0.8:3002"}
catalog-service-1 |
{"level":30,"time":1778821314899,"pid":1,"hostname":"6fe30a59ec47","msg":"catalog-service rodando na porta 3002"}
catalog-service-1 |
{"level":30,"time":1778823411549,"pid":1,"hostname":"6fe30a59ec47","reqId":"req-1","req":{"method":"GET","url":"/catalog","host":"catalog-service:3002","remoteAddress":"172.18.0.9","remotePort":59502},"msg":"incoming request"}
catalog-service-1 |
{"level":30,"time":1778823411561,"pid":1,"hostname":"6fe30a59ec47","reqId":"req-1","res":{"statusCode":200},"responseTime":11.298262999858707,"msg":"request completed"}
catalog-service-1 |
{"level":30,"time":1778824751581,"pid":1,"hostname":"6fe30a59ec47","reqId":"req-2","req":{"method":"GET","url":"/catalog/1/license","host":"catalog-service:3002","remoteAddress":"172.18.0.6","remotePort":52916},"msg":"incoming request"}
gateway-1 | "remotePort": 59132
gateway-1 | }
gateway-1 | [05:16:03] INFO (1): Route GET:/favicon.ico not found
gateway-1 | reqId: "req-2"
gateway-1 | [05:16:03] INFO (1): request completed
gateway-1 | reqId: "req-2"
gateway-1 | res: {
gateway-1 | "statusCode": 404
gateway-1 | }
gateway-1 | responseTime: 0.5451549999415874
gateway-1 | [05:28:33] INFO (1): incoming request
gateway-1 | reqId: "req-3"
gateway-1 | req: {
gateway-1 | "method": "POST",
gateway-1 | "url": "/auth/login",
gateway-1 | "host": "localhost:8080",
gateway-1 | "remoteAddress": "172.18.0.1",
gateway-1 | "remotePort": 49190
catalog-service-1 |
{"level":30,"time":1778824751584,"pid":1,"hostname":"6fe30a59ec47","reqId":"req-2","res":{"statusCode":200},"responseTime":2.6079330001957715,"msg":"request completed"}

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catalog-service-1         |
{"level":30,"time":1778824822784,"pid":1,"hostname":"6fe30a59ec47","reqId":"req-3","req":{"
method":"GET","url":"/health","host":"catalog-service:3002","remoteAddress":"172.18.0.9","re
motePort":34234},"msg":"incoming request"}
gateway-1                 |     }
gateway-1                 | [05:28:33] INFO (1): fetching from remote server
gateway-1                 |     reqId: "req-3"
gateway-1                 |     source: "/login"
gateway-1                 | [05:28:33] INFO (1): response received
gateway-1                 |     reqId: "req-3"
gateway-1                 | [05:28:33] INFO (1): request completed
gateway-1                 |     reqId: "req-3"
gateway-1                 |     res: {
gateway-1                 |       "statusCode": 200
gateway-1                 |     }
gateway-1                 |     responseTime: 116.90150800021365
gateway-1                 | [05:35:51] INFO (1): incoming request
gateway-1                 |     reqId: "req-4"
gateway-1                 |     req: {
gateway-1                 |       "method": "POST",
gateway-1                 |       "url": "/auth/login",
gateway-1                 |       "host": "localhost:8080",
gateway-1                 |       "remoteAddress": "172.18.0.1",
gateway-1                 |       "remotePort": 41782
gateway-1                 |     }
gateway-1                 | [05:35:51] INFO (1): fetching from remote server
gateway-1                 |     reqId: "req-4"
gateway-1                 |     source: "/login"
gateway-1                 | [05:35:51] INFO (1): response received
gateway-1                 |     reqId: "req-4"
gateway-1                 | [05:35:51] INFO (1): request completed
gateway-1                 |     reqId: "req-4"
gateway-1                 |     res: {
gateway-1                 |       "statusCode": 200
gateway-1                 |     }
gateway-1                 |     responseTime: 62.74550499999896
gateway-1                 | [05:36:32] INFO (1): incoming request
gateway-1                 |     reqId: "req-5"
gateway-1                 |     req: {
gateway-1                 |       "method": "POST",
gateway-1                 |       "url": "/auth/login",
gateway-1                 |       "host": "localhost:8080",
gateway-1                 |       "remoteAddress": "172.18.0.1",
gateway-1                 |       "remotePort": 59318
gateway-1                 |     }
gateway-1                 | [05:36:32] INFO (1): fetching from remote server
gateway-1                 |     reqId: "req-5"
gateway-1                 |     source: "/login"
```

gateway-1 | [05:36:32] INFO (1): response received
notification-service-1 |
{"level":30,"time":1778821313961,"pid":1,"hostname":"b4a5d8231a2e","msg":"Server listening at http://127.0.0.1:3005"}
notification-service-1 |
{"level":30,"time":1778821313962,"pid":1,"hostname":"b4a5d8231a2e","msg":"Server listening at http://172.18.0.2:3005"}
notification-service-1 |
{"level":30,"time":1778821313962,"pid":1,"hostname":"b4a5d8231a2e","msg":"notification-service rodando na porta 3005"}
notification-service-1 |
{"level":30,"time":1778824751747,"pid":1,"hostname":"b4a5d8231a2e","reqId":"req-1","req":{"method":"POST","url":"/notify","host":"notification-service:3005","remoteAddress":"172.18.0.6","remotePort":58888},"msg":"incoming request"}
notification-service-1 |
{"level":30,"time":1778824752037,"pid":1,"hostname":"b4a5d8231a2e","userId":"user_1","type":"session_started","delay":273,"msg":"Notificação enviada (simulada em 273ms)"}
notification-service-1 |
{"level":30,"time":1778824752050,"pid":1,"hostname":"b4a5d8231a2e","reqId":"req-1","res":{"statusCode":200},"responseTime":299.72768699983135,"msg":"request completed"}
notification-service-1 |
{"level":30,"time":1778824822800,"pid":1,"hostname":"b4a5d8231a2e","reqId":"req-2","req":{"method":"GET","url":"/health","host":"notification-service:3005","remoteAddress":"172.18.0.9","remotePort":51106},"msg":"incoming request"}
notification-service-1 |
{"level":30,"time":1778824822801,"pid":1,"hostname":"b4a5d8231a2e","reqId":"req-2","res":{"statusCode":200},"responseTime":0.6854990003630519,"msg":"request completed"}
auth-service-1 | Seed: 3 usuários criados
auth-service-1 |
{"level":30,"time":1778821314563,"pid":1,"hostname":"c276d5cdb3da","msg":"Server listening at http://127.0.0.1:3001"}
auth-service-1 |
{"level":30,"time":1778821314563,"pid":1,"hostname":"c276d5cdb3da","msg":"Server listening at http://172.18.0.3:3001"}
auth-service-1 |
{"level":30,"time":1778821314563,"pid":1,"hostname":"c276d5cdb3da","msg":"auth-service rodando na porta 3001"}
auth-service-1 |
{"level":30,"time":1778822913506,"pid":1,"hostname":"c276d5cdb3da","reqId":"req-1","req":{"method":"POST","url":"/login","host":"auth-service:3001","remoteAddress":"172.18.0.9","remotePort":43966},"msg":"incoming request"}
auth-service-1 |
{"level":30,"time":1778822913582,"pid":1,"hostname":"c276d5cdb3da","reqId":"req-1","res":{"statusCode":200},"responseTime":75.22205300000496,"msg":"request completed"}
auth-service-1 |
{"level":30,"time":1778823351341,"pid":1,"hostname":"c276d5cdb3da","reqId":"req-2","req":{"method":"POST","url":"/login","host":"auth-service:3001","remoteAddress":"172.18.0.9","remotePort":38368},"msg":"incoming request"}

auth-service-1 |
{"level":30,"time":1778823351399,"pid":1,"hostname":"c276d5cdb3da","reqId":"req-2","res":{"
statusCode":200},"responseTime":57.69680300028995,"msg":"request completed"}
auth-service-1 |
{"level":30,"time":1778823392115,"pid":1,"hostname":"c276d5cdb3da","reqId":"req-3","req":{"
method":"POST","url":"/login","host":"auth-service:3001","remoteAddress":"172.18.0.9","rem
otePort":38368},"msg":"incoming request"}
auth-service-1 |
{"level":30,"time":1778823392178,"pid":1,"hostname":"c276d5cdb3da","reqId":"req-3","res":{"
statusCode":200},"responseTime":62.68488800013438,"msg":"request completed"}
auth-service-1 |
{"level":30,"time":1778824822770,"pid":1,"hostname":"c276d5cdb3da","reqId":"req-4","req":{"
method":"GET","url":"/health","host":"auth-service:3001","remoteAddress":"172.18.0.9","remo
tePort":59420},"msg":"incoming request"}
auth-service-1 |
{"level":30,"time":1778824822771,"pid":1,"hostname":"c276d5cdb3da","reqId":"req-4","res":{"
statusCode":200},"responseTime":0.8585379999130964,"msg":"request completed"}
streaming-service-1 |
{"level":30,"time":1778821314476,"pid":1,"hostname":"de2eee6858d3","msg":"Server
listening at http://127.0.0.1:3003"}
streaming-service-1 |
{"level":30,"time":1778821314477,"pid":1,"hostname":"de2eee6858d3","msg":"Server
listening at http://172.18.0.6:3003"}
streaming-service-1 |
{"level":30,"time":1778821314477,"pid":1,"hostname":"de2eee6858d3","msg":"streaming-ser
vice rodando na porta 3003"}
streaming-service-1 |
{"level":30,"time":1778824751487,"pid":1,"hostname":"de2eee6858d3","reqId":"req-1","req":{
"method":"POST","url":"/streaming/play","host":"streaming-service:3003","remoteAddress":"1
72.18.0.9","remotePort":49442},"msg":"incoming request"}
streaming-service-1 |
{"level":30,"time":1778824751503,"pid":1,"hostname":"de2eee6858d3","movieId":1,"userId":"
user_1","msg":"Iniciando fluxo de play..."}
streaming-service-1 |
{"level":30,"time":1778824752079,"pid":1,"hostname":"de2eee6858d3","reqId":"req-1","res":{"
statusCode":201},"responseTime":587.4574139998294,"msg":"request completed"}
streaming-service-1 |
{"level":30,"time":1778824822789,"pid":1,"hostname":"de2eee6858d3","reqId":"req-2","req":{
"method":"GET","url":"/health","host":"streaming-service:3003","remoteAddress":"172.18.0.9"
,"remotePort":41154},"msg":"incoming request"}
streaming-service-1 |
{"level":30,"time":1778824822791,"pid":1,"hostname":"de2eee6858d3","reqId":"req-2","res":{"
statusCode":200},"responseTime":1.2073380001820624,"msg":"request completed"}
catalog-service-1 |
{"level":30,"time":1778824822785,"pid":1,"hostname":"6fe30a59ec47","reqId":"req-3","res":{"
statusCode":200},"responseTime":0.7890599998645484,"msg":"request completed"}
gateway-1 | reqId: "req-5"
gateway-1 | [05:36:32] INFO (1): request completed

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gateway-1          |    reqId: "req-5"
gateway-1          |    res: {
gateway-1          |      "statusCode": 200
gateway-1          |    }
gateway-1          |    responseTime: 65.03108400013298
gateway-1          | [05:36:51] INFO (1): incoming request
gateway-1          |    reqId: "req-6"
gateway-1          |    req: {
gateway-1          |      "method": "GET",
gateway-1          |      "url": "/api/catalog",
gateway-1          |      "host": "localhost:8080",
gateway-1          |      "remoteAddress": "172.18.0.1",
gateway-1          |      "remotePort": 48056
gateway-1          |    }
gateway-1          | [05:36:51] INFO (1): fetching from remote server
gateway-1          |    reqId: "req-6"
gateway-1          |    source: "/catalog"
gateway-1          | [05:36:51] INFO (1): response received
gateway-1          |    reqId: "req-6"
gateway-1          | [05:36:51] INFO (1): request completed
gateway-1          |    reqId: "req-6"
gateway-1          |    res: {
gateway-1          |      "statusCode": 200
gateway-1          |    }
gateway-1          |    responseTime: 18.776246999856085
gateway-1          | [05:59:11] INFO (1): incoming request
gateway-1          |    reqId: "req-7"
gateway-1          |    req: {
gateway-1          |      "method": "POST",
gateway-1          |      "url": "/api/streaming/play",
gateway-1          |      "host": "localhost:8080",
gateway-1          |      "remoteAddress": "172.18.0.1",
gateway-1          |      "remotePort": 49350
gateway-1          |    }
gateway-1          | [05:59:11] INFO (1): fetching from remote server
gateway-1          |    reqId: "req-7"
gateway-1          |    source: "/streaming/play"
gateway-1          | [05:59:12] INFO (1): response received
gateway-1          |    reqId: "req-7"
gateway-1          | [05:59:12] INFO (1): request completed
gateway-1          |    reqId: "req-7"
gateway-1          |    res: {
gateway-1          |      "statusCode": 201
gateway-1          |    }
gateway-1          |    responseTime: 647.741280999966
gateway-1          | [05:59:31] INFO (1): incoming request
gateway-1          |    reqId: "req-8"
gateway-1          |    req: {
```
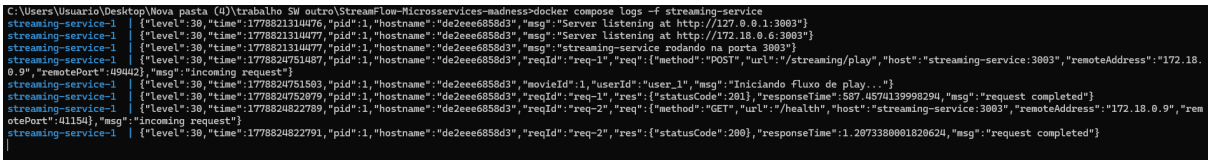

```
gateway-1          |      "method": "GET",
gateway-1          |      "url": "/api/recommendations/user_1",
gateway-1          |      "host": "localhost:8080",
gateway-1          |      "remoteAddress": "172.18.0.1",
gateway-1          |      "remotePort": 35344
gateway-1          |    }
gateway-1          | [05:59:31] INFO (1): fetching from remote server
gateway-1          |    reqId: "req-8"
gateway-1          |    source: "/recommendations/user_1"
gateway-1          | [05:59:31] INFO (1): response received
gateway-1          |    reqId: "req-8"
gateway-1          | [05:59:31] INFO (1): request completed
gateway-1          |    reqId: "req-8"
gateway-1          |    res: {
gateway-1          |      "statusCode": 200
gateway-1          |    }
gateway-1          |    responseTime: 8.58689699973911
gateway-1          | [05:59:47] INFO (1): incoming request
gateway-1          |    reqId: "req-9"
gateway-1          |    req: {
gateway-1          |      "method": "GET",
gateway-1          |      "url": "/api/billing/user_1",
gateway-1          |      "host": "localhost:8080",
gateway-1          |      "remoteAddress": "172.18.0.1",
gateway-1          |      "remotePort": 47106
gateway-1          |    }
gateway-1          | [05:59:47] INFO (1): fetching from remote server
gateway-1          |    reqId: "req-9"
gateway-1          |    source: "/billing/user_1"
gateway-1          | [05:59:47] INFO (1): response received
gateway-1          |    reqId: "req-9"
gateway-1          | [05:59:47] INFO (1): request completed
gateway-1          |    reqId: "req-9"
gateway-1          |    res: {
gateway-1          |      "statusCode": 200
gateway-1          |    }
gateway-1          |    responseTime: 60.744218999985605
gateway-1          | [06:00:00] INFO (1): incoming request
gateway-1          |    reqId: "req-a"
gateway-1          |    req: {
gateway-1          |      "method": "GET",
gateway-1          |      "url": "/api/analytics/report",
gateway-1          |      "host": "localhost:8080",
gateway-1          |      "remoteAddress": "172.18.0.1",
gateway-1          |      "remotePort": 53020
gateway-1          |    }
gateway-1          | [06:00:00] INFO (1): fetching from remote server
gateway-1          |    reqId: "req-a"
```

```
gateway-1          |    source: "/analytics/report"
gateway-1          | [06:00:01] INFO (1): response received
gateway-1          |    reqId: "req-a"
gateway-1          | [06:00:01] INFO (1): request completed
gateway-1          |    reqId: "req-a"
gateway-1          |    res: {
gateway-1          |      "statusCode": 200
gateway-1          |    }
gateway-1          |    responseTime: 333.61655399994925
gateway-1          | [06:00:22] INFO (1): incoming request
gateway-1          |    reqId: "req-b"
gateway-1          |    req: {
gateway-1          |      "method": "GET",
gateway-1          |      "url": "/health",
gateway-1          |      "host": "localhost:8080",
gateway-1          |      "remoteAddress": "172.18.0.1",
gateway-1          |      "remotePort": 48258
gateway-1          |    }
gateway-1          | [06:00:22] INFO (1): request completed
gateway-1          |    reqId: "req-b"
gateway-1          |    res: {
gateway-1          |      "statusCode": 503
gateway-1          |    }
gateway-1          |    responseTime: 58.48715800000355
```

## Ver logs de um serviço específico

docker compose logs -f streaming-service




C:\Users\Usuario\Desktop\Nova pasta (4)\trabalho SW outro\StreamFlow-Microsservices-madness>docker compose logs -f streaming-service
streaming-service-1 | {"level":30,"time":1778821314476,"pid":1,"hostname":"de2eee6858d3","msg":"Server listening at http://127.0.0.1:3003"}
streaming-service-1 | {"level":30,"time":1778821314477,"pid":1,"hostname":"de2eee6858d3","msg":"Server listening at http://172.18.0.6:3003"}
streaming-service-1 | {"level":30,"time":1778821314477,"pid":1,"hostname":"de2eee6858d3","msg":"streaming-service rodando na porta 3003"}

streaming-service-1 |
{"level":30,"time":1778824751487,"pid":1,"hostname":"de2eee6858d3","reqId":"req-1","req":{"method":"POST","url":"/streaming/play","host":"streaming-service:3003","remoteAddress":"172.18.0.9","remotePort":49442},"msg":"incoming request"}
streaming-service-1 |
{"level":30,"time":1778824751503,"pid":1,"hostname":"de2eee6858d3","movieId":1,"userId":"user_1","msg":"Iniciando fluxo de play..."}
streaming-service-1 |
{"level":30,"time":1778824752079,"pid":1,"hostname":"de2eee6858d3","reqId":"req-1","res":{"statusCode":201},"responseTime":587.4574139998294,"msg":"request completed"}
streaming-service-1 |
{"level":30,"time":1778824822789,"pid":1,"hostname":"de2eee6858d3","reqId":"req-2","req":{"method":"GET","url":"/health","host":"streaming-service:3003","remoteAddress":"172.18.0.9","remotePort":41154},"msg":"incoming request"}
streaming-service-1 |
{"level":30,"time":1778824822791,"pid":1,"hostname":"de2eee6858d3","reqId":"req-2","res":{"statusCode":200},"responseTime":1.2073380001820624,"msg":"request completed"}

docker compose logs -f gateway

```
C:\Users\Usuario\Desktop\Nova pasta (4)\trabalho SW outro\StreamFlow-Microsservices-madness>docker compose logs -f gateway
gateway-1  | [05:01:55] INFO (1): Server listening at http://127.0.0.1:8080
gateway-1  | [05:01:55] INFO (1): Server listening at http://172.18.0.9:8080
gateway-1  | [05:01:55] INFO (1): StreamFlow Gateway operando na porta 8080
gateway-1  | [05:16:03] INFO (1): incoming request
gateway-1  |     reqId: "req-1"
gateway-1  |     req: {
gateway-1  |       "method": "GET",
gateway-1  |       "url": "/",
gateway-1  |       "host": "localhost:8080",
gateway-1  |       "remoteAddress": "172.18.0.1",
gateway-1  |       "remotePort": 59132
gateway-1  |     }
gateway-1  | [05:16:03] INFO (1): Route GET:/ not found
gateway-1  |     reqId: "req-1"
gateway-1  | [05:16:03] INFO (1): request completed
gateway-1  |     reqId: "req-1"
gateway-1  |     res: {
gateway-1  |       "statusCode": 404
gateway-1  |     }
gateway-1  |     responseTime: 25.292331999866292
gateway-1  | [05:16:03] INFO (1): incoming request
gateway-1  |     reqId: "req-2"
gateway-1  |     req: {
gateway-1  |       "method": "GET",
```

```
gateway-1  |      "url": "/favicon.ico",
gateway-1  |      "host": "localhost:8080",
gateway-1  |      "remoteAddress": "172.18.0.1",
gateway-1  |      "remotePort": 59132
gateway-1  |    }
gateway-1  | [05:16:03] INFO (1): Route GET:/favicon.ico not found
gateway-1  |    reqId: "req-2"
gateway-1  | [05:16:03] INFO (1): request completed
gateway-1  |    reqId: "req-2"
gateway-1  |    res: {
gateway-1  |      "statusCode": 404
gateway-1  |    }
gateway-1  |    responseTime: 0.5451549999415874
gateway-1  | [05:28:33] INFO (1): incoming request
gateway-1  |    reqId: "req-3"
gateway-1  |    req: {
gateway-1  |      "method": "POST",
gateway-1  |      "url": "/auth/login",
gateway-1  |      "host": "localhost:8080",
gateway-1  |      "remoteAddress": "172.18.0.1",
gateway-1  |      "remotePort": 49190
gateway-1  |    }
gateway-1  | [05:28:33] INFO (1): fetching from remote server
gateway-1  |    reqId: "req-3"
gateway-1  |    source: "/login"
gateway-1  | [05:28:33] INFO (1): response received
gateway-1  |    reqId: "req-3"
gateway-1  | [05:28:33] INFO (1): request completed
gateway-1  |    reqId: "req-3"
gateway-1  |    res: {
gateway-1  |      "statusCode": 200
gateway-1  |    }
gateway-1  |    responseTime: 116.90150800021365
gateway-1  | [05:35:51] INFO (1): incoming request
gateway-1  |    reqId: "req-4"
gateway-1  |    req: {
gateway-1  |      "method": "POST",
gateway-1  |      "url": "/auth/login",
gateway-1  |      "host": "localhost:8080",
gateway-1  |      "remoteAddress": "172.18.0.1",
gateway-1  |      "remotePort": 41782
gateway-1  |    }
gateway-1  | [05:35:51] INFO (1): fetching from remote server
gateway-1  |    reqId: "req-4"
gateway-1  |    source: "/login"
gateway-1  | [05:35:51] INFO (1): response received
gateway-1  |    reqId: "req-4"
gateway-1  | [05:35:51] INFO (1): request completed
```

```
gateway-1  |     reqId: "req-4"
gateway-1  |     res: {
gateway-1  |       "statusCode": 200
gateway-1  |     }
gateway-1  |     responseTime: 62.74550499999896
gateway-1  | [05:36:32] INFO (1): incoming request
gateway-1  |     reqId: "req-5"
gateway-1  |     req: {
gateway-1  |       "method": "POST",
gateway-1  |       "url": "/auth/login",
gateway-1  |       "host": "localhost:8080",
gateway-1  |       "remoteAddress": "172.18.0.1",
gateway-1  |       "remotePort": 59318
gateway-1  |     }
gateway-1  | [05:36:32] INFO (1): fetching from remote server
gateway-1  |     reqId: "req-5"
gateway-1  |     source: "/login"
gateway-1  | [05:36:32] INFO (1): response received
gateway-1  |     reqId: "req-5"
gateway-1  | [05:36:32] INFO (1): request completed
gateway-1  |     reqId: "req-5"
gateway-1  |     res: {
gateway-1  |       "statusCode": 200
gateway-1  |     }
gateway-1  |     responseTime: 65.03108400013298
gateway-1  | [05:36:51] INFO (1): incoming request
gateway-1  |     reqId: "req-6"
gateway-1  |     req: {
gateway-1  |       "method": "GET",
gateway-1  |       "url": "/api/catalog",
gateway-1  |       "host": "localhost:8080",
gateway-1  |       "remoteAddress": "172.18.0.1",
gateway-1  |       "remotePort": 48056
gateway-1  |     }
gateway-1  | [05:36:51] INFO (1): fetching from remote server
gateway-1  |     reqId: "req-6"
gateway-1  |     source: "/catalog"
gateway-1  | [05:36:51] INFO (1): response received
gateway-1  |     reqId: "req-6"
gateway-1  | [05:36:51] INFO (1): request completed
gateway-1  |     reqId: "req-6"
gateway-1  |     res: {
gateway-1  |       "statusCode": 200
gateway-1  |     }
gateway-1  |     responseTime: 18.776246999856085
gateway-1  | [05:59:11] INFO (1): incoming request
gateway-1  |     reqId: "req-7"
gateway-1  |     req: {
```

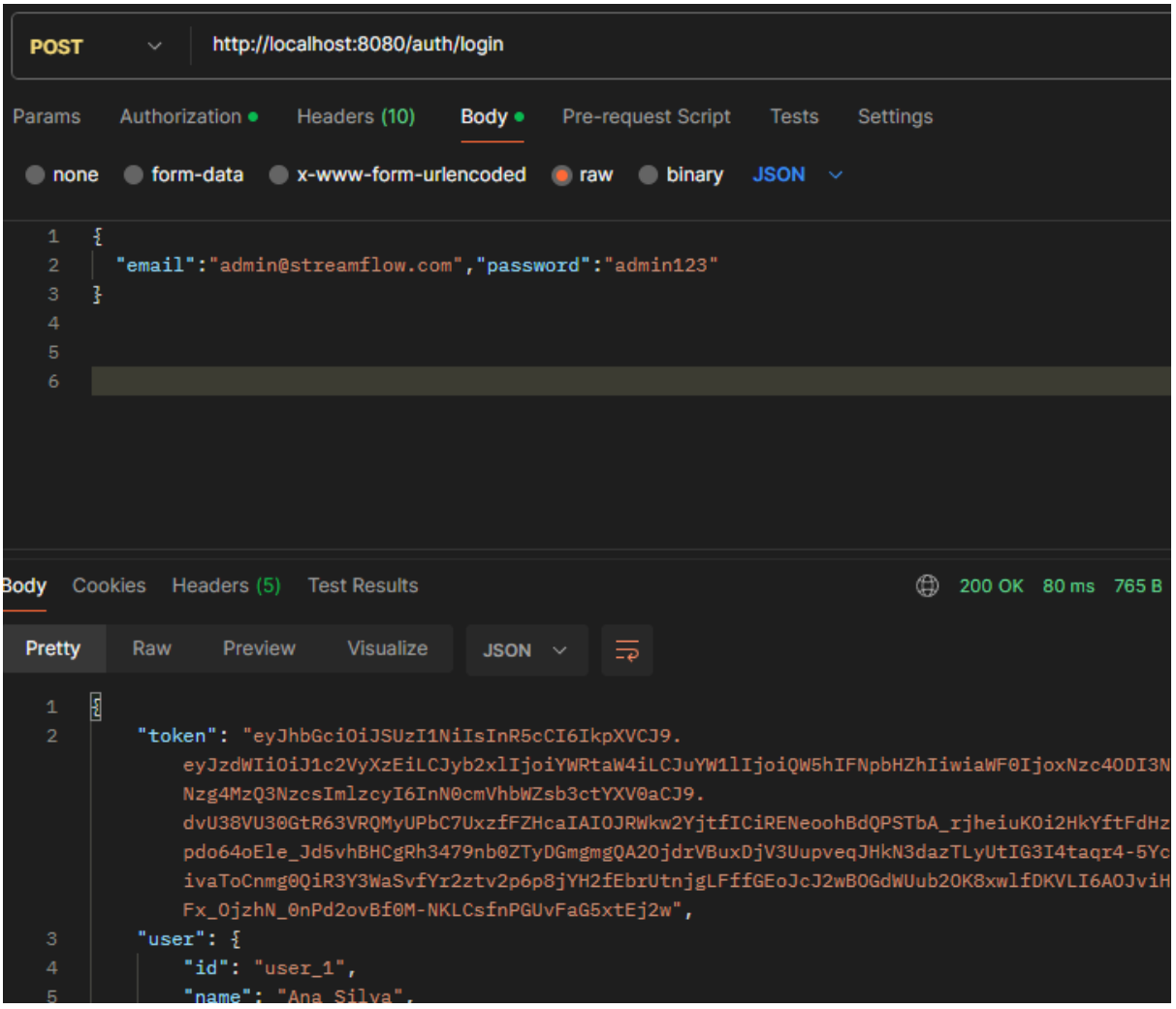
```
gateway-1  |      "method": "POST",
gateway-1  |      "url": "/api/streaming/play",
gateway-1  |      "host": "localhost:8080",
gateway-1  |      "remoteAddress": "172.18.0.1",
gateway-1  |      "remotePort": 49350
gateway-1  |    }
gateway-1  | [05:59:11] INFO (1): fetching from remote server
gateway-1  |    reqId: "req-7"
gateway-1  |    source: "/streaming/play"
gateway-1  | [05:59:12] INFO (1): response received
gateway-1  |    reqId: "req-7"
gateway-1  | [05:59:12] INFO (1): request completed
gateway-1  |    reqId: "req-7"
gateway-1  |    res: {
gateway-1  |      "statusCode": 201
gateway-1  |    }
gateway-1  |    responseTime: 647.741280999966
gateway-1  | [05:59:31] INFO (1): incoming request
gateway-1  |    reqId: "req-8"
gateway-1  |    req: {
gateway-1  |      "method": "GET",
gateway-1  |      "url": "/api/recommendations/user_1",
gateway-1  |      "host": "localhost:8080",
gateway-1  |      "remoteAddress": "172.18.0.1",
gateway-1  |      "remotePort": 35344
gateway-1  |    }
gateway-1  | [05:59:31] INFO (1): fetching from remote server
gateway-1  |    reqId: "req-8"
gateway-1  |    source: "/recommendations/user_1"
gateway-1  | [05:59:31] INFO (1): response received
gateway-1  |    reqId: "req-8"
gateway-1  | [05:59:31] INFO (1): request completed
gateway-1  |    reqId: "req-8"
gateway-1  |    res: {
gateway-1  |      "statusCode": 200
gateway-1  |    }
gateway-1  |    responseTime: 8.58689699973911
gateway-1  | [05:59:47] INFO (1): incoming request
gateway-1  |    reqId: "req-9"
gateway-1  |    req: {
gateway-1  |      "method": "GET",
gateway-1  |      "url": "/api/billing/user_1",
gateway-1  |      "host": "localhost:8080",
gateway-1  |      "remoteAddress": "172.18.0.1",
gateway-1  |      "remotePort": 47106
gateway-1  |    }
gateway-1  | [05:59:47] INFO (1): fetching from remote server
gateway-1  |    reqId: "req-9"
```

```
gateway-1  |     source: "/billing/user_1"
gateway-1  | [05:59:47] INFO (1): response received
gateway-1  |     reqId: "req-9"
gateway-1  | [05:59:47] INFO (1): request completed
gateway-1  |     reqId: "req-9"
gateway-1  |     res: {
gateway-1  |       "statusCode": 200
gateway-1  |     }
gateway-1  |     responseTime: 60.744218999985605
gateway-1  | [06:00:00] INFO (1): incoming request
gateway-1  |     reqId: "req-a"
gateway-1  |     req: {
gateway-1  |       "method": "GET",
gateway-1  |       "url": "/api/analytics/report",
gateway-1  |       "host": "localhost:8080",
gateway-1  |       "remoteAddress": "172.18.0.1",
gateway-1  |       "remotePort": 53020
gateway-1  |     }
gateway-1  | [06:00:00] INFO (1): fetching from remote server
gateway-1  |     reqId: "req-a"
gateway-1  |     source: "/analytics/report"
gateway-1  | [06:00:01] INFO (1): response received
gateway-1  |     reqId: "req-a"
gateway-1  | [06:00:01] INFO (1): request completed
gateway-1  |     reqId: "req-a"
gateway-1  |     res: {
gateway-1  |       "statusCode": 200
gateway-1  |     }
gateway-1  |     responseTime: 333.61655399994925
gateway-1  | [06:00:22] INFO (1): incoming request
gateway-1  |     reqId: "req-b"
gateway-1  |     req: {
gateway-1  |       "method": "GET",
gateway-1  |       "url": "/health",
gateway-1  |       "host": "localhost:8080",
gateway-1  |       "remoteAddress": "172.18.0.1",
gateway-1  |       "remotePort": 48258
gateway-1  |     }
gateway-1  | [06:00:22] INFO (1): request completed
gateway-1  |     reqId: "req-b"
gateway-1  |     res: {
gateway-1  |       "statusCode": 503
gateway-1  |     }
gateway-1  |     responseTime: 58.48715800000355
```

## Filtrar logs por nível

Os logs do Fastify (Pino) são estruturados em JSON. Para filtrar erros:

docker compose logs -f | grep '"level":50'



GET http://localhost:8080/api/catalog

Params | Authorization | Headers (10) | Body | Pre-request Script | Tests | Settings

Type: Bearer ...

Token: eyJhbGciOiJSUzI1NiIsInR5

The authorization header will be automatically generated when you send the request.
Learn more about authorization ↗

Store your secrets with end-to-end encryption locally using Vault.

Body | Cookies | Headers (5) | Test Results

200 OK 14 ms 1.8

Pretty | Raw | Preview | Visualize | JSON


```
[
    {
        "id": 1,
        "title": "O Poderoso Chefão",
        "genre": "Drama",
        "year": 1972,
        "rating": 9.2,
        "license_active": 1,
        "synopsis": "A saga da família Corleone no submundo do crime organizado."
    },
```

POST http://localhost:8080/api/streaming/play

Params Authorization Headers (10) Body Pre-request Script Tests Settings

none form-data x-www-form-urlencoded raw binary JSON

```
1 {"movieId": 1}
```

Body Cookies Headers (5) Test Results 201 Created 435 ms 274 B

Pretty Raw Preview Visualize JSON

```
1 {
2     "sessionId": 2,
3     "movie": "O Poderoso Chefão",
4     "status": "playing",
5     "message": "Reprodução iniciada."
6 }
```

GET http://localhost:8080/api/recommendations/user_1

Params Authorization Headers (10) Body Pre-request Script Tests Settings

none form-data x-www-form-urlencoded raw binary JSON

```
1 {"movieId": 1}
```

Body Cookies Headers (5) Test Results 200 OK 34 ms 330 B

Pretty Raw Preview Visualize JSON


```
1  {
2      "userId": "user_1",
3      "preferences": [
4          "Drama",
5          "Ação",
6          "Ficção Científica"
7      ],
8      "recentlyViewed": [
9          1,
10         1
11     ],
```

GET http://localhost:8080/api/billing/user_1

Params Authorization Headers (10) Body Pre-request Script Tests Settings

none form-data x-www-form-urlencoded raw binary JSON

```
{"movieId": 1}
```

Body Cookies Headers (5) Test Results 200 OK 59 ms 540 B

Pretty Raw Preview Visualize JSON


```
{
    "subscription": {
        "id": 1,
        "user_id": "user_1",
        "plan": "Premium",
        "price_monthly": 49.9,
        "status": "active",
        "started_at": "2026-05-15 05:01:54"
    },
```

ttp://localhost:8080/api/analytics/report

http://localhost:8080/api/analytics/report

s Authorization Headers (10) Body Pre-request Script Tests Settings

ne form-data x-www-form-urlencoded raw binary JSON

```
{"movieId": 1}
```

Cookies Headers (5) Test Results Status: 200 OK Time: 324 ms Size: 607 B

Raw Preview Visualize JSON


```
{
    "generatedAt": "2026-05-15T06:53:04.990Z",
    "summary": {
        "totalSubscribers": 3,
        "activeSubscribers": 3,
        "monthlyRecurringRevenue": "104.70",
        "totalInvoices": 5,
        "paidInvoices": 2,
        "pendingInvoices": 3,
        "collectionRate": "40.0%"
    },
```

GET http://localhost:8080/health

Params Authorization Headers (10) Body Pre-request Script Tests Settings

none form-data x-www-form-urlencoded raw binary JSON

```
1 {"movieId": 1}
```

Body Cookies Headers (5) Test Results Status: 503 Service Unavailable Time: 72 ms Size: 1.15 KB

Pretty Raw Preview Visualize JSON


```
1 {
2     "service": "gateway",
3     "status": "degraded",
4     "timestamp": "2026-05-15T06:56:32.159Z",
5     "downstream": {
6         "auth-service": {
7             "status": "ok",
8             "service": "auth-service",
9             "uptime": 6897.547287477,
10            "timestamp": "2026-05-15T06:56:32.100Z"
```


```
lucas@DESKTOP-ON1RKEE:/mnt/c/Users/Usuario/Desktop/Nova pasta (4)/trabalho SW outro/StreamFlow-Microservices-madness$ docker compose logs -f | grep '"level":50'
```

Fazendo os testes de login, mostrar catalogo, iniciar reprodução, recomendações, faturamento e relatório de teste não aparece nenhum erro e logo não foi capturado pelo comando do docker. No comando para ver  saúde geral apareceu um erro de 503 service unavailable mas mesmo assim não foi capturado pelo comando. O comando grep '"level":50' procura por erros fatais ou exceções não tratadas. Ele simplesmente fez um check-up nos outros serviços e decidiu retornar um status 503 porque algo não está 100%. Para o sistema de log, isso é uma operação normal de verificação, por isso ele provavelmente registrou como level: 30 (INFO) ou level: 40 (WARN), e não como 50.

O gateway apareceu degraded.

Geralmente, isso acontece por um desses motivos:

- Serviço Oculto: Pode haver um serviço (como um Banco de Dados ou um Redis) que não apareceu nessa lista mas que o Gateway não conseguiu alcançar.
- Latência: Um dos serviços demorou demais para responder (timeout), e o Gateway marcou como degradado.
- Dependência Externa: O sistema pode estar tentando validar algo fora do Docker que falhou.

docker compose logs -f | grep '"level":40'

Depois de fazer os testes com esse comando tive os mesmos resultados. Por fim fiz um teste com o comando 30.

docker compose logs -f | grep '"level":30'

Resumo dos resultados.

1. Inicialização dos Serviços
Todos os serviços iniciaram corretamente e estão ouvindo em suas respectivas portas:

Auth: Porta 3001

Catalog: Porta 3002

Streaming: Porta 3003

Recommendation: Porta 3004

Notification: Porta 3005

Billing: Porta 3006

Analytics: Porta 3007

2. Atividades de Destaque por Serviço
Auth Service: Processou múltiplos logins com sucesso (HTTP 200). O tempo de resposta variou entre 57ms e 89ms, o que é bem rápido.

Catalog Service: Entregou a lista de filmes (/catalog) e verificou licenças de filmes específicos para o serviço de streaming.

Streaming Service: É o serviço que mais trabalha. Ele iniciou fluxos de "play" para o movieId: 1 (O Poderoso Chefão) e teve um tempo de resposta maior (~1.5 segundos em uma das chamadas), simulando o carregamento do vídeo.

Analytics Service: Executou um "Pipeline de relatório" em 3 etapas (Coleta -> Agregação -> Formatação). Cada relatório levou cerca de 290ms para ser gerado.

Notification Service: Enviou notificações simuladas do tipo session_started para a usuária Ana Silva (user_1).

3. Saúde do Sistema (Health Checks)
Você pode ver várias chamadas para a rota /health em todos os serviços.

Status: Todas retornaram statusCode: 200.

Conclusão: Individualmente, todos os serviços estão saudáveis. Se o seu Gateway diz "degraded", o problema não é a queda de um serviço, mas talvez a comunicação entre eles ou um atraso na resposta que o Gateway recebeu.

4. Fluxo de Integração
Os logs mostram os serviços conversando. Por exemplo:

O Streaming recebe um pedido de play.

Ele chama o Catalog para ver a licença.

Ele avisa o Notification para disparar um aviso de sessão iniciada.

O Recommendation registra que o filme foi visualizado.

## Inspecionar o banco de dados

Para acessar o SQLite dentro de um container:

docker compose exec catalog-service sh

apt update && apt install sqlite3 -y

Tive que alterar o comando pois estou usando ubuntu.

```
ers\Usuario\Desktop\Nova pasta (4)\trabalho SW outro\StreamFlow-Microservices-madness>docker compose exec catalog-
ce sh
update && apt install sqlite3 -y
http://deb.debian.org/debian bookworm InRelease
http://deb.debian.org/debian bookworm-updates InRelease
http://deb.debian.org/debian-security bookworm-security InRelease
ng package lists... Done
ing dependency tree... Done
ng state information... Done
ackages are up to date.
ng package lists... Done
ing dependency tree... Done
ng state information... Done
e3 is already the newest version (3.40.1-2+deb12u2).
raded, 0 newly installed, 0 to remove and 0 not upgraded.
```

sqlite3 /app/data/catalog.db

```
sqlite3 /app/data/catalog.db
Lite version 3.40.1 2022-12-28 14:03:47
ter ".help" for usage hints.
lite>
```

.tables

```
sqlite> .tables
movies
sqlite>
```

SELECT * FROM movies;

```
sqlite> SELECT * FROM movies;
1|O Poderoso Chefão|Drama|1972|9.2|1|A saga da família Corleone no submundo do crime organizado.
2|Interestelar|Ficção Científica|2014|8.7|1|Astronautas viajam por um buraco de minhoca em busca de um novo lar.
3|Parasita|Thriller|2019|8.5|1|Duas famílias de classes sociais opostas se cruzam de forma inesperada.
4|Matrix|Ação|1999|8.7|1|Um hacker descobre que a realidade é uma simulação computacional.
5|Breaking Bad|Drama|2008|9.5|1|Um professor de química se torna produtor de metanfetamina.
6|Stranger Things|Ficção Científica|2016|8.7|1|Crianças enfrentam fenômenos sobrenaturais em uma pequena cidade.
7|Dark|Mistério|2017|8.8|1|Quatro famílias interligadas desvendam uma conspiração temporal.
8|La Casa de Papel|Ação|2017|8.2|0|Um grupo de assaltantes executa um plano no Banco da Espanha.
9|O Gambito da Rainha|Drama|2020|8.6|1|Uma órfã prodígio ascende no mundo do xadrez competitivo.
10|Oppenheimer|Biografia|2023|8.3|1|A história do físico que liderou o Projeto Manhattan.
```

.quit

exit

```
sqlite> .quit
# exit
```

## Cenários de teste para entender os antipatterns

Cenário 1: Cadeia síncrona — impacto de latência

Observe o tempo de resposta do endpoint de play:

```
time curl -s -X POST http://localhost:8080/api/streaming/play \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $TOKEN" \
  -d '{"movieId": 2}'
```

```
lucas@DESKTOP-ON1RKEE:/mnt/c/Users/Usuario/Desktop/Nova pasta (4)/trabalho SW
outro/StreamFlow-Microsservices-madness$ time curl -s -X POST
http://localhost:8080/api/streaming/play \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $TOKEN" \
  -d '{"movieId": 2}'
{"error":"Token expirado."}
real    0m0.115s
```

user  0m0.002s
sys  0m0.011s

```
lucas@DESKTOP-ON1RKEE:/mnt/c/Users/Usuario/Desktop/Nova pasta (4)/trabalho SW outro/StreamFlow-Microsservices-madness$ time curl -s -X POST http://localhost:8080/api/streaming/pl
ay \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $TOKEN" \
  -d '{"movieId": 2}'
{"error":"Token expirado."}
real    0m0.115s
user    0m0.002s
sys     0m0.011s
```

Compare com o tempo de uma chamada direta ao catálogo (rota simples):

```
time curl -s http://localhost:8080/api/catalog \
  -H "Authorization: Bearer $TOKEN"
```

```
lucas@DESKTOP-ON1RKEE:/mnt/c/Users/Usuario/Desktop/Nova pasta (4)/trabalho SW
outro/StreamFlow-Microsservices-madness$ time curl -s http://localhost:8080/api/catalog \
  -H "Authorization: Bearer $TOKEN"
{"error":"Token expirado."}
real    0m0.011s
user    0m0.004s
sys     0m0.002s
```

```
lucas@DESKTOP-ON1RKEE:/mnt/c/Users/Usuario/Desktop/Nova pasta (4)/trabalho SW outro/StreamFlow-Microsservices-madness$ time curl -s http://localhost:8080/api/catalog \
  -H "Authorization: Bearer $TOKEN"
{"error":"Token expirado."}
real    0m0.011s
user    0m0.004s
sys     0m0.002s
```

A diferença de tempo revela a latência acumulada da cadeia síncrona.

Cenário 2: Notification como ponto de falha

Pare o notification-service e observe o que acontece com o play:

```
docker compose stop notification-service
```

```
DESKTOP-ON1RKEE:/mnt/c/Users/Usuario/Desktop/Nova pasta (4)/trabalho SW outro/StreamFlow-Microsservices-madness$ docker compose stop notification-service
top 1/1
ntainer streamflow-microsservices-madness-notification-service-1 Stopped
```

```
curl -s -X POST http://localhost:8080/api/streaming/play \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $TOKEN" \
  -d '{"movieId": 3}' | jq
```

```
lucas@DESKTOP-ON1RKEE:/mnt/c/Users/Usuario/Desktop/Nova pasta (4)/trabalho SW outro/StreamFlow-Microsservices-madness$ curl -s -X POST http://localhost:8080/api/streaming/play \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $TOKEN" \
  -d '{"movieId": 3}' | jq
{
  "sessionId": 4,
  "movie": "Parasita",
  "status": "playing",
  "message": "Reprodução iniciada."
}
```

O play ainda funciona (fail silencioso), mas observe nos logs do streaming-service a mensagem de warning. Em um cenário real com timeout menor, o usuário ficaria esperando até o timeout estourar.

docker compose start notification-service

```
lucas@DESKTOP-ON1RKEE:/mnt/c/Users/Usuario/Desktop/Nova pasta (4)/trabalho SW outro/StreamFlow-Microsservices-madness$ docker compose start notification-service
[+] start 1/1
 ✔Container streamflow-microsservices-madness-notification-service-1 Started
```

## Cenário 3: Banco compartilhado — acoplamento visível

Pare o billing-service e tente gerar um relatório de analytics:

```
lucas@DESKTOP-ON1RKEE:/mnt/c/Users/Usuario/Desktop/Nova pasta (4)/trabalho SW outro/StreamFlow-Microsservices-madness$ docker compose stop billing-service
[+] stop 1/1
 ✔Container streamflow-microsservices-madness-billing-service-1 Stopped
```

curl -s http://localhost:8080/api/analytics/report \
  -H "Authorization: Bearer $TOKEN" | jq

lucas@DESKTOP-ON1RKEE:/mnt/c/Users/Usuario/Desktop/Nova pasta (4)/trabalho SW outro/StreamFlow-Microsservices-madness$ curl -s http://localhost:8080/api/analytics/report \
  -H "Authorization: Bearer $TOKEN" | jq


```
{
  "generatedAt": "2026-05-15T09:48:49.902Z",
  "summary": {
    "totalSubscribers": 3,
    "activeSubscribers": 3,
    "monthlyRecurringRevenue": "104.70",
    "totalInvoices": 5,
    "paidInvoices": 2,
    "pendingInvoices": 3,
    "collectionRate": "40.0%"
  },
  "planDistribution": {
    "Premium": 1,
    "Básico": 1,
    "Standard": 1
  },
  "pipeline": {
```

"note": "Este relatório foi gerado por um pipeline sequencial interno.",
"stages": [
"coleta",
"agregação",
"formatação"
],
"totalProcessingTime": "~250ms"
}
}

```
lucas@DESKTOP-ON1RKEE:/mnt/c/Users/Usuario/Desktop/Nova pasta (4)/trabalho SW outro/StreamFlow-Microsservices-madness$ curl -s http://localhost:8080/api/analytics/report \
  -H "Authorization: Bearer $TOKEN" | jq
{
  "generatedAt": "2026-05-15T09:48:49.902Z",
  "summary": {
    "totalSubscribers": 3,
    "activeSubscribers": 3,
    "monthlyRecurringRevenue": "104.70",
    "totalInvoices": 5,
    "paidInvoices": 2,
    "pendingInvoices": 3,
    "collectionRate": "40.0%"
  },
  "planDistribution": {
    "Premium": 1,
    "Básico": 1,
    "Standard": 1
  },
  "pipeline": {
    "note": "Este relatório foi gerado por um pipeline sequencial interno.",
    "stages": [
      "coleta",
      "agregação",
      "formatação"
    ],
    "totalProcessingTime": "~250ms"
  }
}
```

Isso demonstra o acoplamento: o analytics não depende da API do billing, mas depende do banco do billing. Se o billing alterar o schema da tabela, o analytics quebra sem aviso.

docker compose start billing-service

```
lucas@DESKTOP-ON1RKEE:/mnt/c/Users/Usuario/Desktop/Nova pasta (4)/trabalho SW outro/StreamFlow-Microsservices-madness$ docker compose start billing-service
[+] start 1/1
 ✔Container streamflow-microsservices-madness-billing-service-1 Started
```

## Cenário 4: Health check — visão agregada

curl -s http://localhost:8080/health | jq

lucas@DESKTOP-ON1RKEE:/mnt/c/Users/Usuario/Desktop/Nova pasta (4)/trabalho SW outro/StreamFlow-Microsservices-madness$ curl -s http://localhost:8080/health | jq

{
"service": "gateway",
"status": "degraded",
"timestamp": "2026-05-15T09:52:56.983Z",
"downstream": {
"auth-service": {
"status": "ok",
"service": "auth-service",

```
      "uptime": 17898.235388041,
      "timestamp": "2026-05-15T09:52:56.917Z"
    },
    "catalog-service": {
      "status": "ok",
      "service": "catalog-service",
      "uptime": 17897.832756316,
      "timestamp": "2026-05-15T09:52:56.928Z"
    },
    "streaming-service": {
      "status": "ok",
      "service": "streaming-service",
      "uptime": 17897.867852561,
      "timestamp": "2026-05-15T09:52:56.938Z"
    },
    "recommendation-service": {
      "status": "ok",
      "service": "recommendation-service",
      "uptime": 17897.991759973,
      "timestamp": "2026-05-15T09:52:56.947Z"
    },
    "notification-service": {
      "status": "ok",
      "service": "notification-service",
      "uptime": 455.881808404,
      "timestamp": "2026-05-15T09:52:56.957Z"
    },
    "billing-service": {
      "status": "ok",
      "service": "billing-service",
      "uptime": 110.678196896,
      "timestamp": "2026-05-15T09:52:56.971Z"
    },
    "analytics-service": {
      "status": "ok",
      "service": "analytics-service",
      "uptime": 17897.983577829,
      "timestamp": "2026-05-15T09:52:56.980Z"
    }
  }
}
```

```
lucas@DESKTOP-ON1RKEE:/mnt/c/Users/Usuario/Desktop/Nova pasta (4)/trabalho SW outro/StreamFlow-Microservices-madness$ curl -s http://localhost:8080/health | jq
{
  "service": "gateway",
  "status": "degraded",
  "timestamp": "2026-05-15T09:52:56.983Z",
  "downstream": {
    "auth-service": {
      "status": "ok",
      "service": "auth-service",
      "uptime": 17898.235388041,
      "timestamp": "2026-05-15T09:52:56.917Z"
    },
    "catalog-service": {
      "status": "ok",
      "service": "catalog-service",
      "uptime": 17897.832756316,
      "timestamp": "2026-05-15T09:52:56.928Z"
    },
    "streaming-service": {
      "status": "ok",
      "service": "streaming-service",
      "uptime": 17897.867852561,
      "timestamp": "2026-05-15T09:52:56.938Z"
    },
```